Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#3)

Apply Behavioral-level Verilog to Design and Simulate Combinational Modules

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MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____
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Prepared by:

Verified by: Dr. Nasir Ali Shah Updated: Fall 2026

Task 1:

Description:
In this task I designed a 4-bit adder/subtractor using Verilog behavioral modeling.The circuit takes two 4-bit numbers as inputs.A control signal is used to select addition or subtraction.When the control signal is 0, the circuit performs addition.When the control signal is 1, the circuit performs subtraction.A testbench was used to check the circuit with different input values. Then at end I used pasted code on modelsim and started simulation I got the output diagram.
Code:

```
module AddSub(a,b,ctrl,result);

input [3:0] a;

input [3:0] b;

input ctrl;

output reg [4:0] result;



always @(*)

begin

if (ctrl==0)

result = a+b;

else

result = a-b;

end
```

```
endmodule



module testbench();

reg [3:0] x;

reg [3:0] y;

reg c;

wire [4:0] res;



AddSub uut(x,y,c,res);



initial

begin

x=3; y=1; c=0;

#50

x=7; y=2; c=0;

#50

x=5; y=3; c=1;

#50

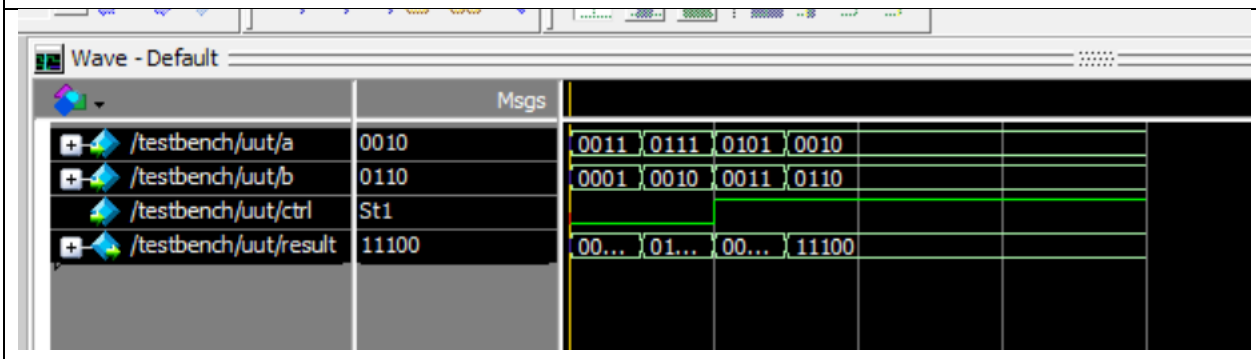
x=2; y=6; c=1;

#50;

end

endmodule
```

Output:


Wave - Default
Msgs
/testbench/uut/a
/testbench/uut/b
/testbench/uut/ctrl
/testbench/uut/result
0010
0110
St1
11100
0011
0111
0101
0010
0001
0010
0011
0110
00...
01...
00...
11100

Task 2:

Description:
In this task we designed a 2×1 MUX using Verilog behavioral modeling. The circuit has two inputs, A and B, and one select input. When the select input is 0, the circuit performs an AND operation on A and B. When the select input is 1, the circuit performs an OR operation on A and B. The output is observed for different combinations of inputs. A testbench was used to verify the working of the circuit. And at the end the code with simulated and we got the output diagram.
Code:

```
module Mux2x1(a,b,sel,y);

input a;

input b;

input sel;

output reg y;



always @(*)

begin

if (sel==0)

y = a&b;

else

y = a|b;

end

endmodule
```

```
module testbench();

reg x;

reg y;

reg s;

wire out;



Mux2x1 uut(x,y,s,out);



initial

begin

x=0; y=0; s=0;

#50

x=0; y=1; s=0;

#50

x=1; y=1; s=0;

#50

x=0; y=1; s=1;

#50

x=1; y=0; s=1;

#50

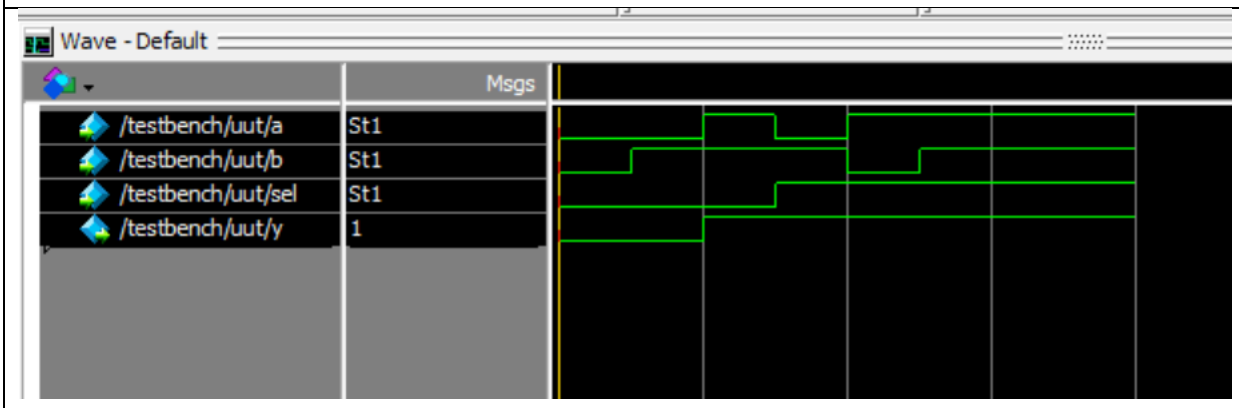
x=1; y=1; s=1;
```

```
#50;

end

endmodule
```

Output:



Task 3:

Description:

In this task I designed a 2-bit comparator using Verilog behavioral modeling. The circuit compares two 2-bit binary numbers, A and B. It then checks whether both numbers are equal, or one is greater or smaller than the other. Three outputs are used to show the comparison result. Only one output becomes 1 according to the comparison. A testbench was used to test the circuit with different input combinations. And at last I placed code on multisim and simulated it and got the output diagram.

Code:

```
module Comparator(a,b,eq,gt,lt);

input [1:0] a;

input [1:0] b;

output reg eq;

output reg gt;

output reg lt;



always @(*)

begin

if (a==b)

begin

eq=1; gt=0; lt=0;

end

else if (a>b)

begin

eq=0; gt=1; lt=0;

end

else

begin

eq=0; gt=0; lt=1;
```

```
end

end

endmodule



module testbench();

reg [1:0] x;

reg [1:0] y;

wire e;

wire g;

wire l;



Comparator uut(x,y,e,g,l);



initial

begin

x=1; y=1;

#50

x=3; y=1;

#50

x=0; y=2;

#50

x=2; y=2;
```

```
#50;

end

endmodule
```

Output:

